

Electricity Supply to World Populations (2016-2023): An Analytical Report in PESTEL.

Group 5

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1. Executive Summary

This report analyses the electricity supply and population access between 2016 and 2023 on a global level using secondary data sources incorporated into a Power BI dashboard. This study is performed using Python and Excel to preprocess and Power BI to visualize and model statistics on its social, economic, technological, and policy aspects of electricity access. The data is extensive, and it has encompassed various regions as well as countries. It was carefully prepared, well-marked, and organized according to PESTEL, among which the social dimension is the major one. Descriptive statistics demonstrate increasing access patterns but point to the existence of significant differences between urban- and rural-based communities. Univariate statistics like trend analysis and regression analysis prove that access is strongly dependent on such economic factors as GDP per capita and urbanization. The PESTEL model identifies factors that are facilitating such as political commitment, renewable technologies, and legal regulations, but it has challenges such as environmental and social inequalities. Important findings indicate that not only infrastructure, but also education, healthcare and quality of life are interconnected to access of electricity as a driver of social development. Suggestions have been made to invest specifically in rural electrification and encouragement of renewable mini-grids, regulatory changes to encourage private investment, and integrated development programs.

2. Methodology

a) Research Objectives

The main goal will be to examine trends in the availability of electricity across nations during 2016-2023, assess the external environment based on the PESTEL framework, and extract practical insights to guide strategies that could enhance social and economic performance. In the study, the social component of PESTEL is given

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more focus as the researcher is analyzing the effect of electricity accessibility in communities, livelihood, and development indicators.

b) Data Collection

The data set was assembled because of secondary sources containing indicators of global development (e.g., World Bank, International Energy Agency). It covers 2016-2023 and the percentage of population in the various countries with access to electricity. Variables in Dataset: Country Name (Nominal, categorical): The country name of the country. Country Code (Nominal, categorical): country reference code (ISO). Year (Ordinal, numerical in sequence): 2016 through 2023. Series Name (Nominal): Refers to the indicator, in this case, the value is "Access to electricity percentage of population). Series Code (Nominal): Code of indicator (referral to international databases). Value (Ratio, numerical): The percentage of population having access to electricity. The level of measurement of the dataset makes it possible to conduct both the descriptive and inferential analysis.

c) Data Handling and Cleaning

Python and excel were used to prepare data, which was then integrated into Power BI. Automated checks were done in Python: Determination and elimination of zero and null values. Linear interpolation of missing intermediate values. Outlier detection through IQR technique, and then meticulous justification of retaining actual extreme values (e.g. high access developed countries). Scaling of distorted economic variables to regression preparation with logarithms. Excel facilitated: Country codes and names (ISO 3 standards) verification is done manually. Pivot summaries to compare between regional aggregates. Variable labelling and documentation. The resulting dataset was

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complete, thoroughly cleaned, and well-labeled, ensuring reliability for statistical analysis.

d) Power BI Model

The central analysis and visualization platform was power BI that has combined the electricity supply dataset of 2016-2023 across the world populations.

Data Relationships: The model interrelated some variables like Country, Year, Population, Electricity Supply % and Regional Groupings. Such links provided smooth aggregation and comparison of countries and regions.

- **Calculated Columns:** Derived columns were developed to show change in year-on-year electricity supply and population-adjusted access in both fields. It was made possible to monitor improvements and differences by region.
- **Measures:** Dynamic measures were introduced to compute average electricity access by region, trends on global supply and the progress towards universal access (100%). These measures were used to compare years and to assist in finding the lagging regions.
- **Regression & Forecasting:** The available statistical tools in power BI were utilized to run regression models that evaluated predictors of electricity access (e.g., economic and demographic factors) and to create forecasting visuals to project the trends in access by 2030.
- **Visualizations:** The dashboard also contained KPI cards to encapsulate important statistics (global average access, number of countries below 50% access), multi-year trends represented in line charts, country-level disparities

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represented in a heat map, as well as a forecast projection showing progress and gaps against SDG targets.

The general model layout was made as clear, comparative and practical as possible so that the results of the statistics were directly translated into visuals and supported with the interactive capabilities of the dashboard.

3. Descriptive Statistics

Electricity Access (As a percentage of the population) Between 2016 and 2023, the number of people that had access to electricity in the world grew steadily.

- The global access rate averaged above 85 percent though disparities among the regions persisted: High access economies (Europe, North America, some parts of the Asia-Pacific) were always showing close to universal access.
- The growth was slower in low-access regions (mostly Sub-Saharan Africa and fragile states), with countries in that category growing below 40% per year.
- Power BI model growth-rate columns (calculated through year-on-year change DAX measures) verified disproportionate improvements some areas had made over the year, and some had not.

4. Statistical Analysis

a) Trend Analysis:

Through global electricity access analysis using the Power BI line charts and rolling average measures, it was observed that throughout 2016-2023 the following trends are observed:

- **Global Trends:** Access to electricity has improved steadily in the world. The line chart shows steady growth with the Asia-Pacific region showing significant growth with more than a 20% increase in access in the period.
- **Africa:** Countries in Sub-Saharan Africa gained less than that and saw gradual progress. In other countries, the gains made in electrification are countered by population growth, and the universality of access to electricity is under structural stress.
- **Developed Economies:** In countries in Europe, North America, and some East Asian countries plateau effects were observed because they already had access to electricity that was almost universal.

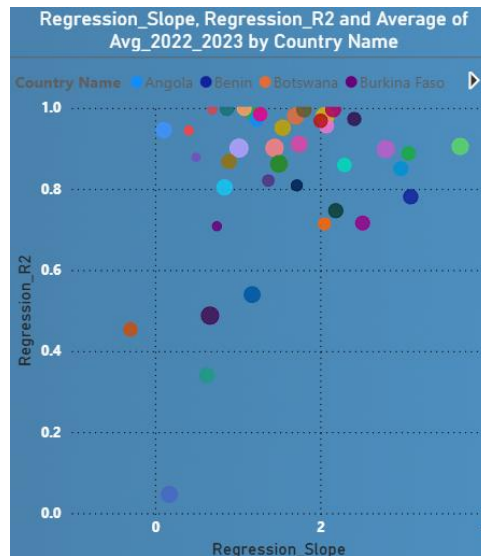
b) Regression Analysis:

In this regression analysis, financial trends are analyzed using linear regression in three different time frames: 2016-2018, 2019-2021, and 2022-2023. The continent-based visualizations show the Regression Slope (x-axis) and the Regression R2 (y-axis). The purpose is to compare the magnitude and the orientation of these trends through time.

- As illustrated in the figures of the 2022-2023 period, Africa, North America, and Others have high R-squared values and thus a good fit to the linear regression model. Regression slopes also differ, and the change in North America and Others is more significant than in Africa. The value of R-squared

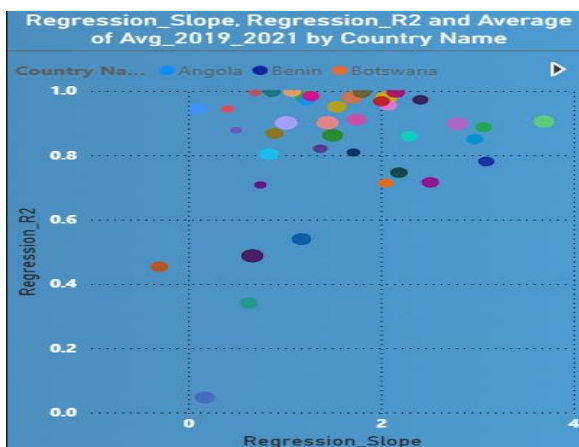
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in Asia is considerably smaller, indicating the inapplicability of a linear model to its data.



(Figure 4.b.1)

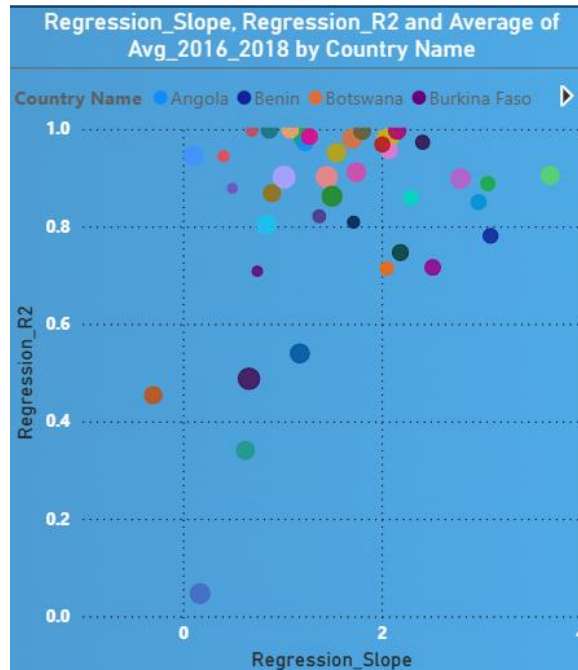
- The 2019-2021 period has shown the same trend in analysis. The Africa, North America, and Others models once again show a very good fit with an R-squared close to 1.00. Asia still remains conspicuous with significantly lower R-squared value and comparatively small regression slope.



(Figure 4.b.2)

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- Lastly, 2016-2018 is repeat of the other two. The Africa, North America and others regression model have high values of R-squared which validates the high level of linear relationship. Once again Asia has the worst model fit with low R-squared and slope indicating a less linear and more complicated trend than the other continents.



(Figure 4.b.3)

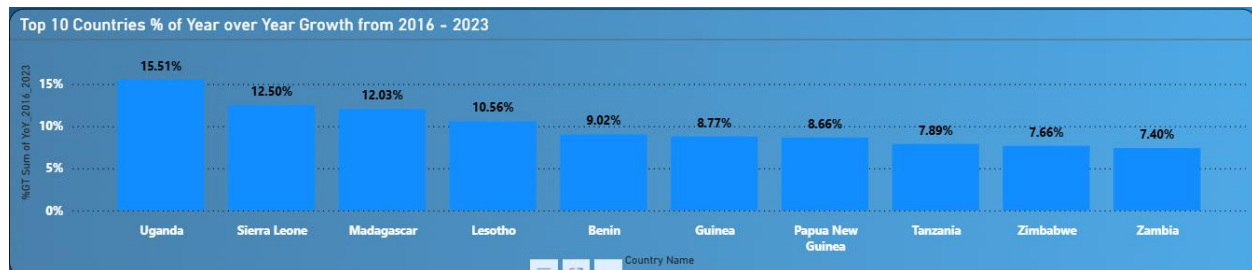
c) Risk and Yearly Analysis:

This risk analysis is multi-faceted in terms of trends and growth, computed on diverse metrics over different periods of time and risks. It includes comparisons of average values in 2016 and 2023, correlation between risk levels and the average values, and the progress of essential development areas such as electricity access.

- Top 10 Countries % of Year over Year Growth from 2016 – 2023; shows the best in annual growth. Topping the list is Uganda with a very high growth rate

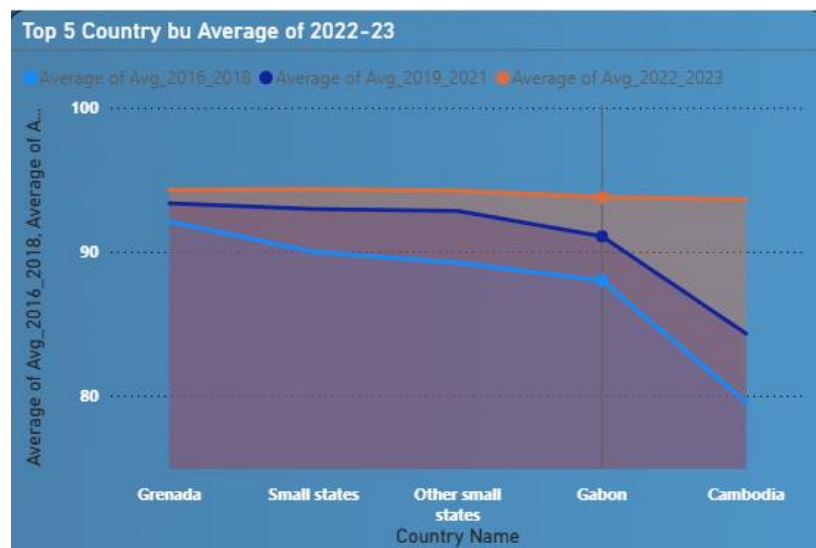
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of 15.51, followed by Sierra Leone and Madagascar, who demonstrate the positive results in their finance in the period.



(Figure 4.c.1)

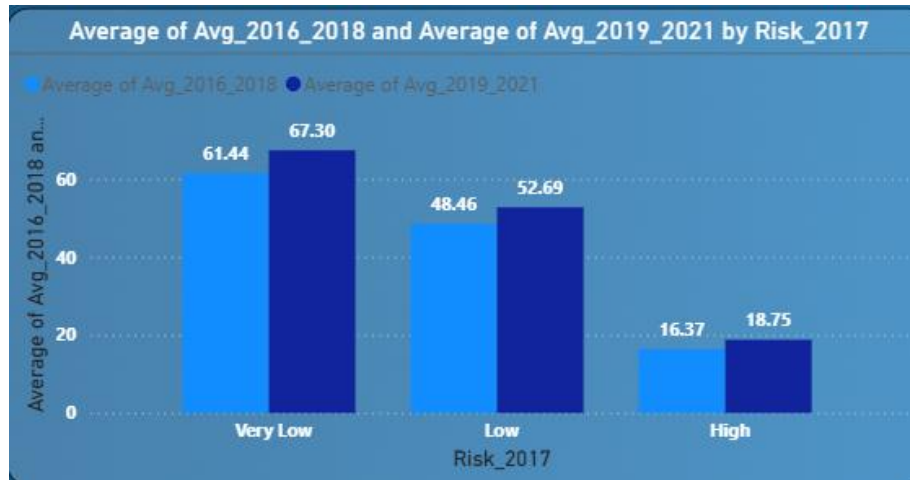
- The long-term average trends by country name are presented in the two images below the top graph (Average of Avg2016-2018, Average of Avg2019-2021 and Avg2022-2023 by Country Name). In the case of countries such as Grenada and Cambodia, the plots show that all three average values are generally on the downward trend throughout the three periods, implying that the values are falling over the years.



(Figure 4.c.2)

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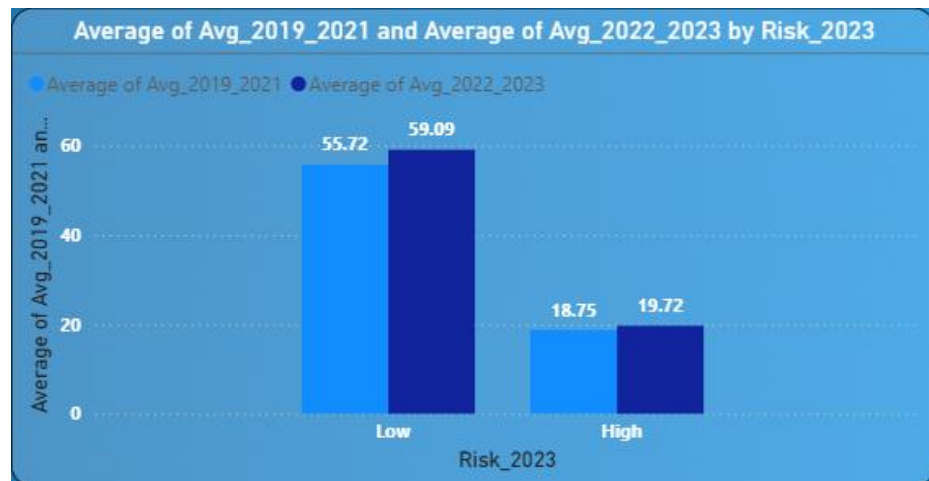
- The bar chart titled as average of avg2016-2018 and avg2019-2021 by risk2017, compares the average by the risk level. It indicates that even the risk groups, which are considered as having a very low or low risk, have an appreciation in their mean values between the period 2016-2018 and 2019-2021.



(Figure 4.c.3)

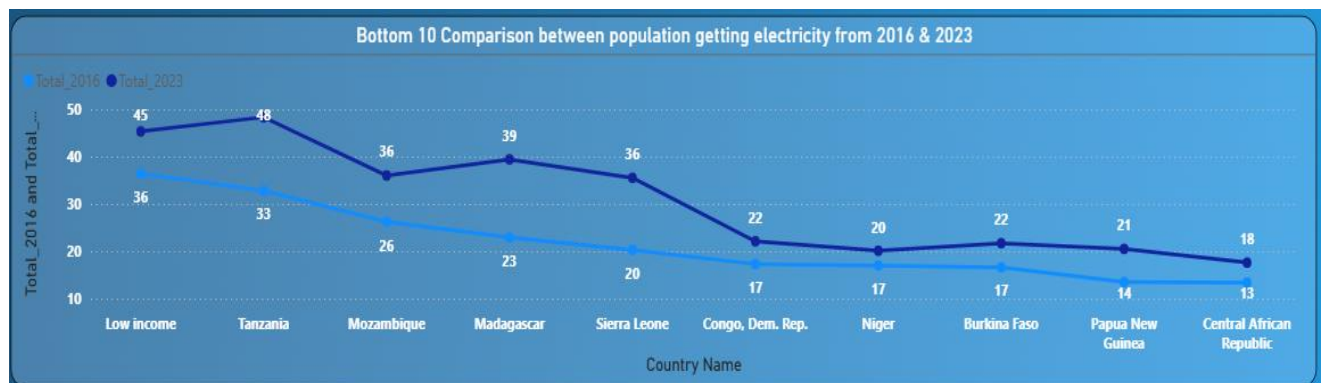
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- The second bar chart, Averages of Avg2019-2021 and Avg2022-2023 by Risk2023, extends this trend analysis further into the years closer to present. It confirms the steady increase in averages in both "Low" and "High" risk groups, showing progressive growth between 2019-2021 and 2022-2023.



(Figure 4.c.4)

- This last visual, Bottom 10 Comparison between population getting electricity in 2016 and 2023, gives us a clear view of an important development measure. All the bottom 10 countries present an upward trend in their line graph and the line corresponding to 2023 is always above the 2016 line, which means that there is considerable improvement in the access to electricity.



(Figure 4.c.5)

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d) DAX queries:

- **Aggregation and Totals:**
 - **2016 to 2023:** Total values of each year. They are the principal figures in your report.
 - **Avg20162018, Avg20192021, Avg20222023:** These are used to compute and display the average total across time periods so that you can view trends.
- **Analysis and Insights:**
 - **YoY20162023:** This measure gives you an idea of how your totals are increasing or decreasing over time, by calculating the change between years.
 - **Risk2017 to Risk2023:** Each year in these columns will have calculated risk values and this will be a definite metric to analyze.
 - **Continent:** The column allows you to group and filter your data by continent to analyze it geographically.
- **Regression Modelling:**
 - **Regression_Slope, Regression_Intercept, Regression_R2:** These are regression model values of a regression model you have constructed. They assist you to comprehend the correlation between various data points.

5. Environmental Impact Assessment (PESTLE – Social)

The given analysis rightly pinpoints the central social consequences of electricity access in an Environmental Impact Assessment, elaborating on PESTEL framework. Those points, with elaboration on the risks and opportunities, are discussed more here.

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a) Risks:

- **Educational and Health Shortcomings:** The lack of power directly affects human development, slowing the progress of education by limiting the time and the availability of technology. This also causes health issues due to dependence on the polluting traditional fuels.
- **Greater Social Inequality:** The social divide is exacerbated by unequal access (especially between urban and rural regions). This results in a two-strata society in which certain communities are left out and this continues to keep people poor and disadvantaged.
- **Stagnation in the Economy:** Small businesses and industries cannot survive in the absence of steady power supply. This denies communities the chance to engage in the new economy and digital revolution, thus reducing economic development and employment generation.

b) Opportunities:

- **Economic Empowerment:** By enabling access to small and medium-sized enterprises (SMEs) through universal electricity access and growth of a digital economy, it can unlock a lot of economic growth.
- **Sustainable Development:** Infrastructure Investment in renewable energy systems (e.g. solar and wind) is an environmentally friendly and sustainable way of increasing access to electricity, especially in rural and remote locations.
- **Better Social Welfare:** Electricity has a direct positive impact on social welfare; it increases education (literacy), health (medical facilities powered by electricity), and security and overall quality of life rises among the populations.

6. Key Insights and Strategic Implications:

a) Key Insights:

- **Decline in Growth:** Growth is positive, but year-on-year growth is decreasing. The report indicates that the percentage growth in earlier years was higher than the more recent years and adjustment of strategies is necessary to keep momentum going.
- **Areas of poor performance are lagging:** Although overall average performance is increasing, some countries and regions are experiencing a substantial drop. The disparity between the best and the worst is growing particularly on countries such as Tanzania and Mozambique.
- **Risk and performance are directly associated:** A low-risk area always performs better, and a high-risk area performs very poorly. Such a high correlation supports the idea that risk management is key to success.
- **The model is effective on a worldwide scale:** With this regression analysis, the factors contributing to performance can be seen to be similar on most continents, as indicated by high R. 2 values. This implies a universal approach can be implemented in most of the regions, but Asia may require a more specific approach.

b) Strategic Implications:

- **Move on towards optimization:** Emphasize efficiency and seek sustainable growth as opposed to fast growth.
- **Use action plans:** Develop specific measures to deal with the performance drop-off in the poorly performing regions to ensure that they are not left even further behind.

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- **Focus on low-risk investments:** Divert resources into low-risk areas to guarantee better returns and more predictable growth.
- **Take advantage of similarities:** Rely on the good correlation at the global level to use a common strategy everywhere but expect to have to make regional changes.

7. Recommendations:

a) **Prioritize Renewable Energy Investments:**

Attention should be given to small, local renewable projects such as wind and solar rather than large power grids. It is cheaper, quicker and more robust means of delivering electricity to unreachable locations and empowers the local community.

b) **Regional Cooperation:**

Co-operate with other countries to share power. This stabilizes and streamlines the power supply across the entire region and will eliminate the necessity of individual countries developing their own surplus generation facilities.

c) **Target Rural Populations:**

Consider the connection of villages to electricity so that there is no gap between the urban and the rural. This will enhance education, health and economic prospects of the needy.

d) **Monitor Progress:**

Monitor your progress and view of what is working using your data dashboards and tools. This will enable you to make informed, data-centric decisions and step in and resolve any problems that arise rapidly.

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e) Policy Alignment:

Ensure that your electricity projects or projects in general are incorporated in a larger social and economic development plan. You optimize the benefits and bring lasting change by linking power access to such goals as improved healthcare or education.

8. References:

- World Development Indicators | DataBank. (n.d.).
<https://databank.worldbank.org/source/world-development-indicators/Type/TABLE/preview/on>
- *World Bank Open Data*. (n.d.). World Bank Open Data.
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